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Modeling Citation Trustworthiness in Evolving Scholarly Networks

Software Engineering Department | Capstone Project Phase B - 61998 | Project No. 26-1-R-9

Problem Definition

Citation systems are widely used to measure academic impact, but they can be affected by artificial citations, excessive self-citation, forced citations, and citation cartels. The main challenge is evaluating citation trustworthiness in large evolving networks without labeled data.

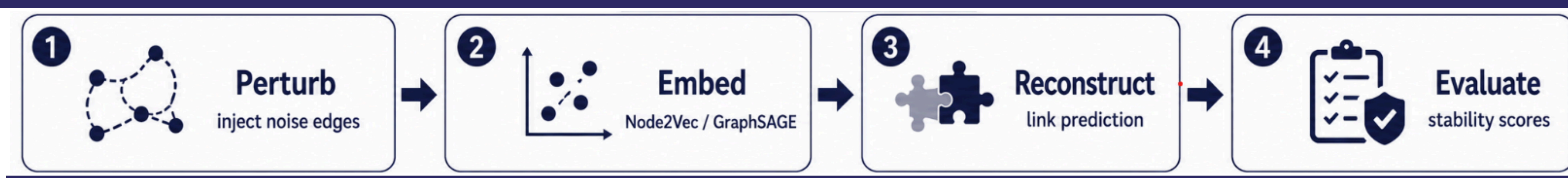
Design Goals

The goal is to build a graph-based framework that detects weak or suspicious citation links using stability scores over time.

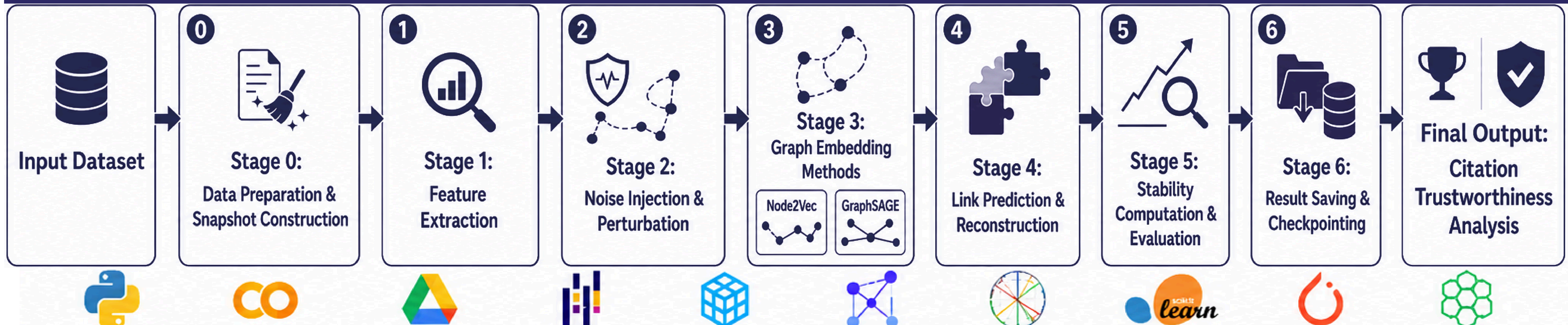
Our Approach

Each paper is represented as a graph node, and each citation is treated as an edge. The system builds yearly citation snapshots, injects noise edges, applies Node2Vec and GraphSAGE embeddings, reconstructs removed links, and computes stability scores.

Conceptual Pipeline

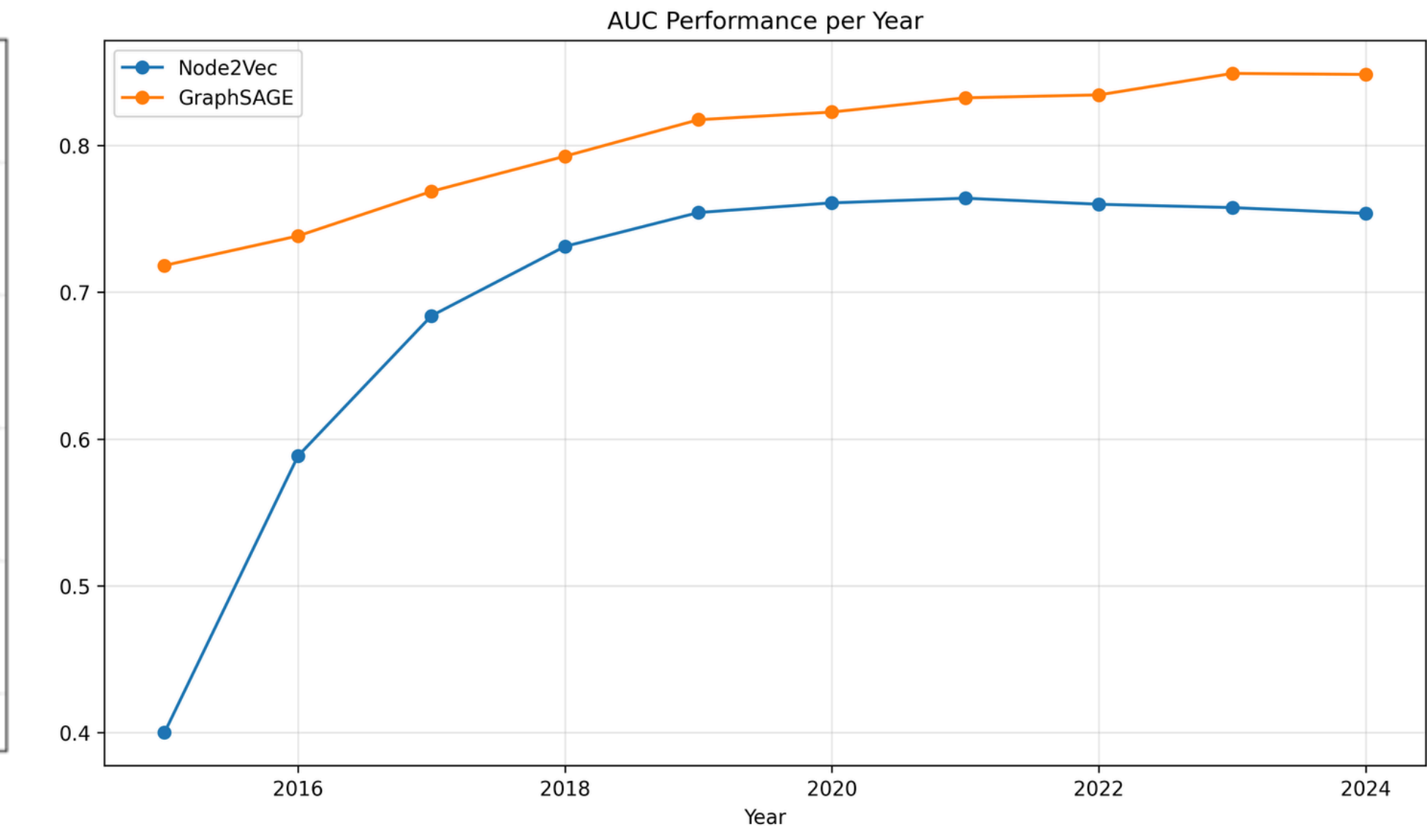
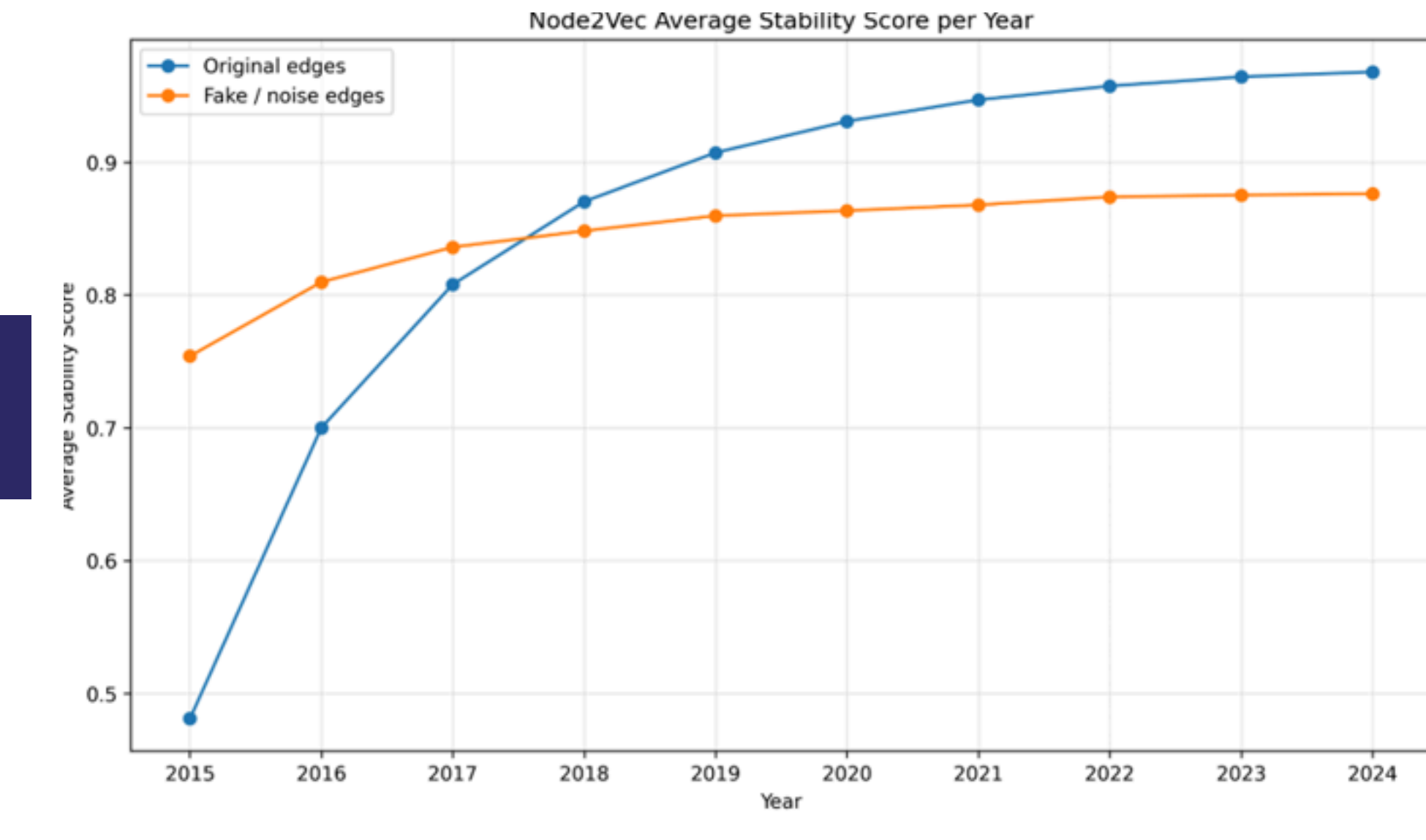
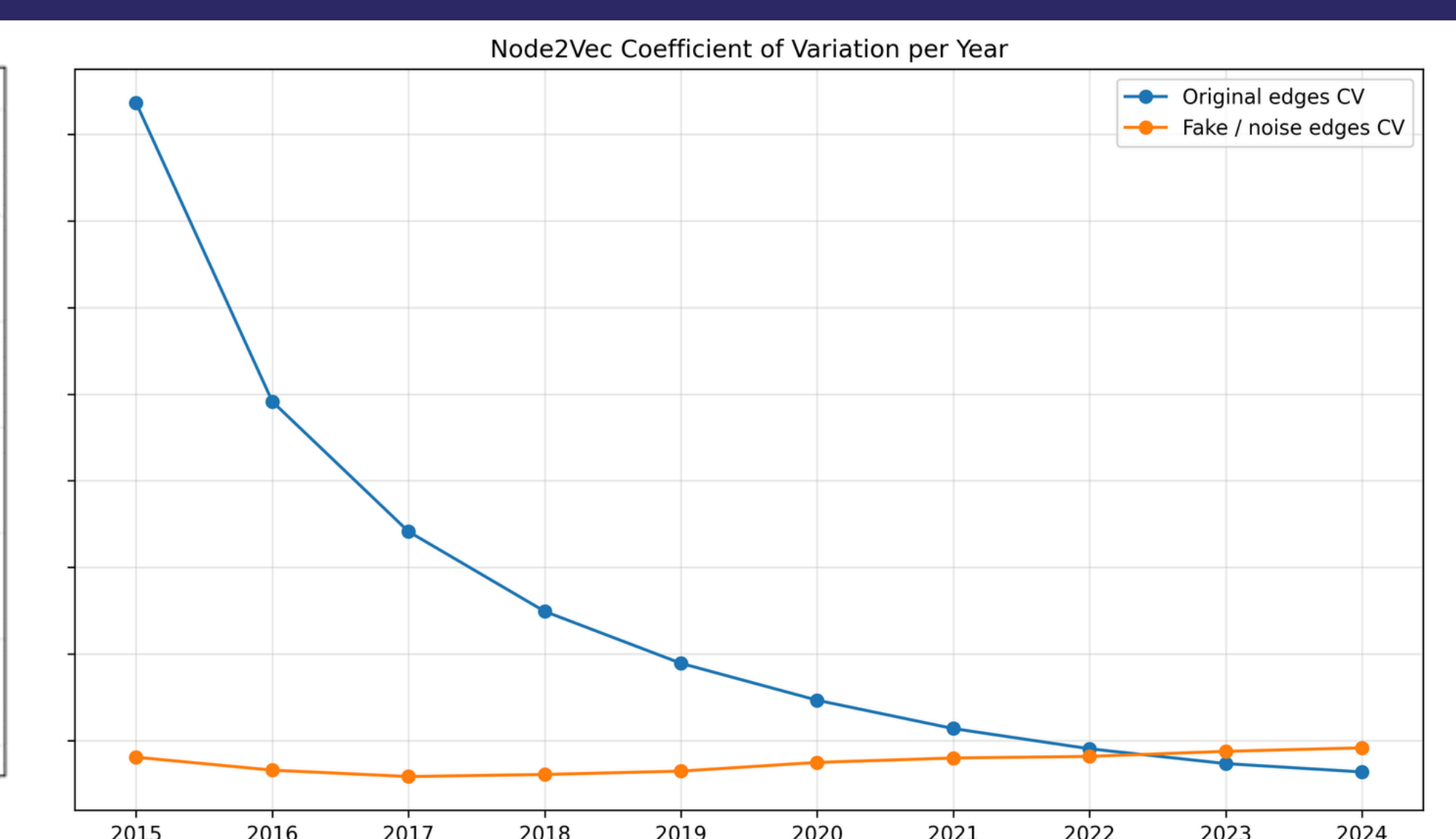
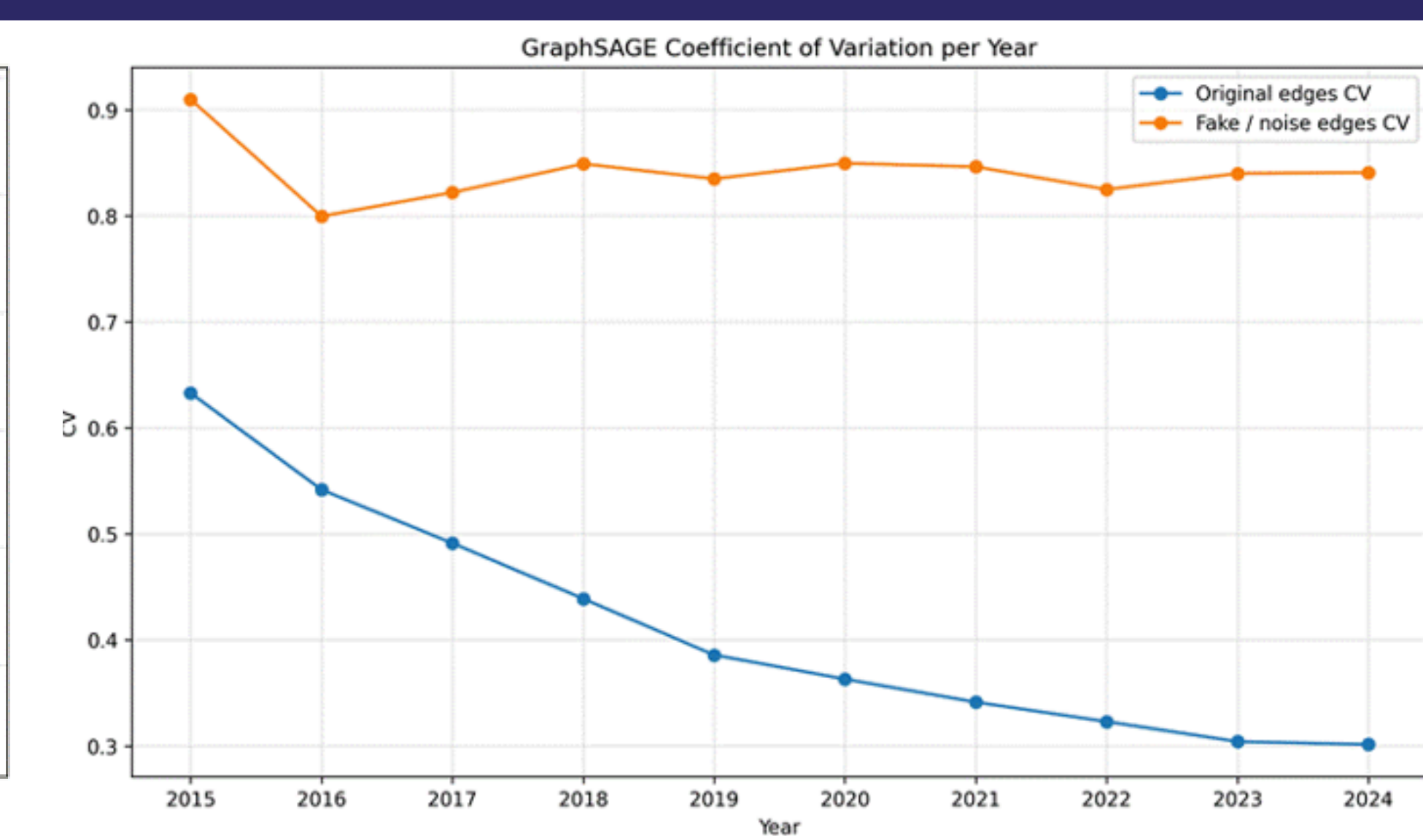
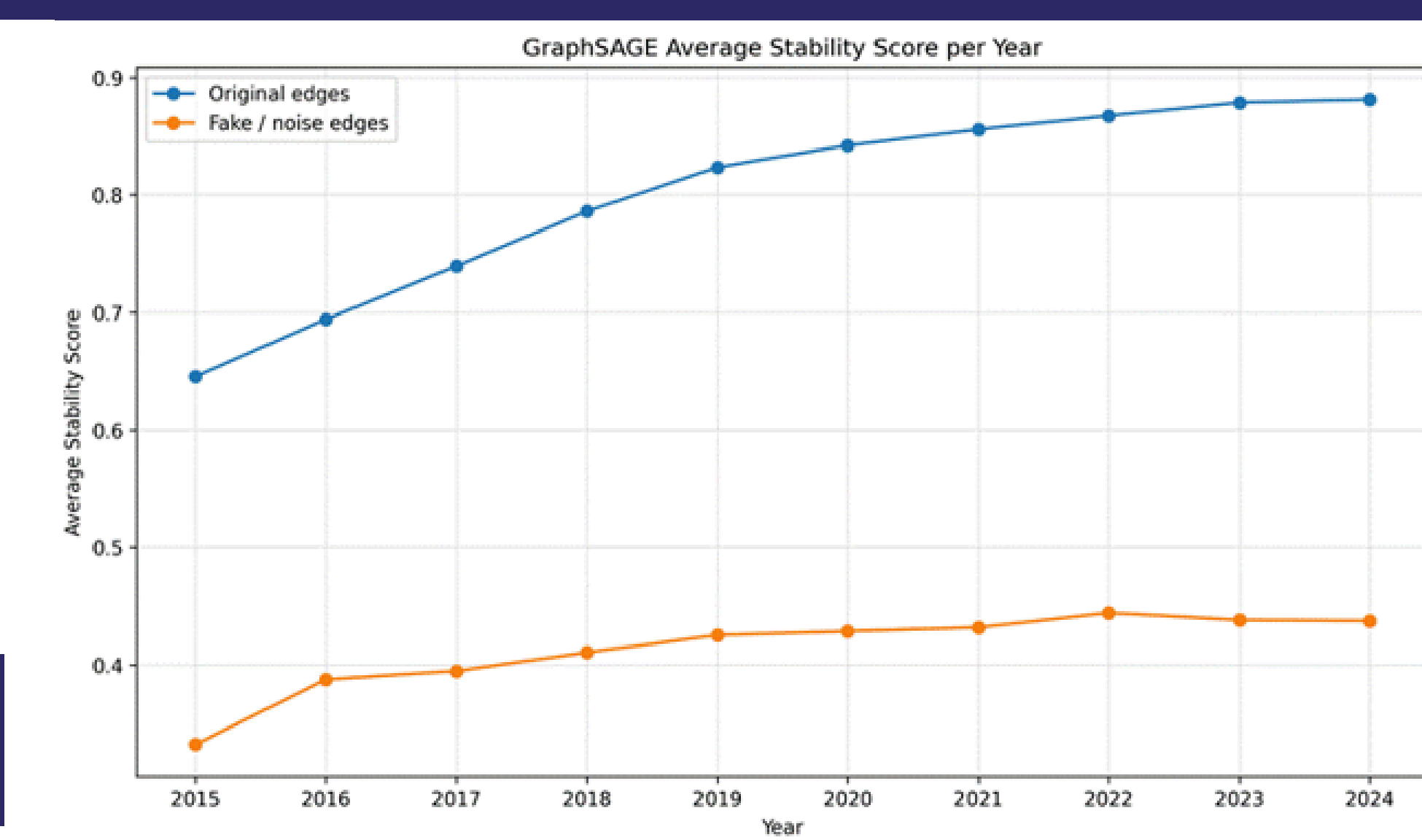


Technical Pipeline

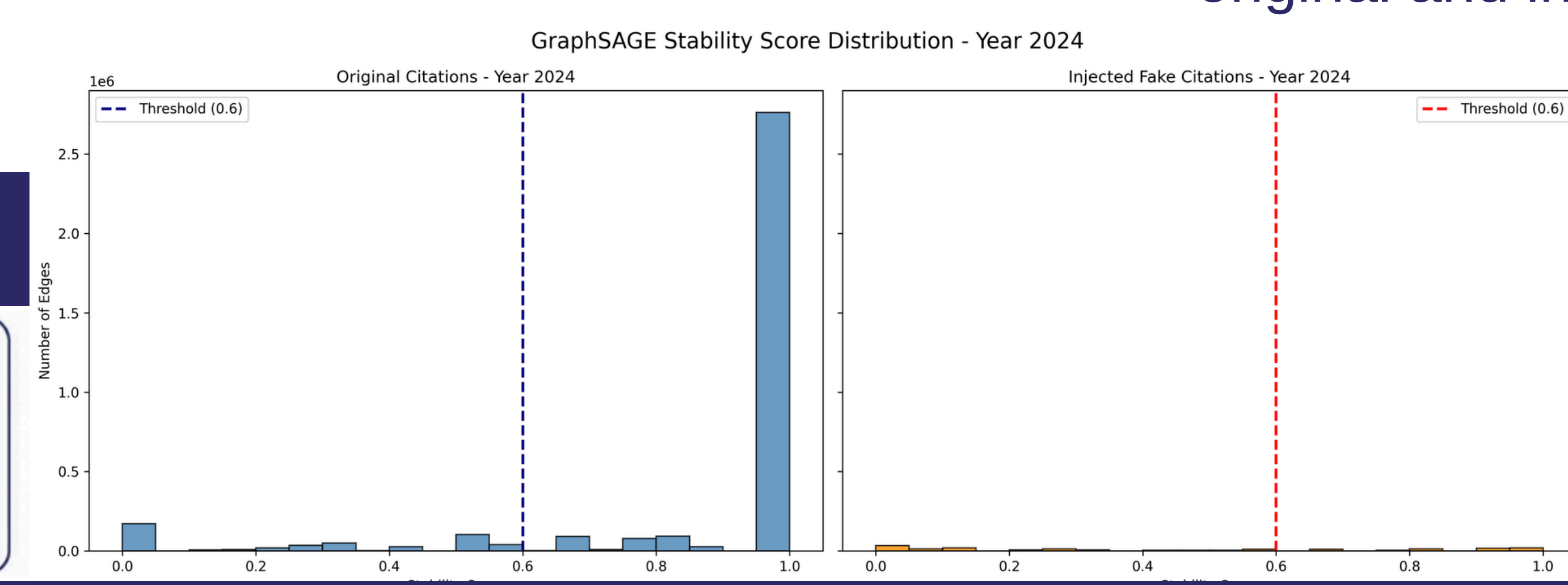


Conclusion

The project demonstrates that graph-based stability analysis can provide useful indicators for citation trustworthiness. Higher stability of original edges suggests stronger structural support, while lower stability of injected noise edges indicates weaker network support. These indicators can help identify citation links that may require further investigation.



The stability score distribution shows that original edges are concentrated at higher stability values, while injected noise edges receive lower scores. The AUC results confirm that GraphSAGE achieved higher and more stable detection performance across the years. GraphSAGE provided clearer separation between original and injected noise edges than Node2Vec.



Original citation edges achieved higher stability scores than injected noise edges, indicating stronger structural support in the citation network.